Phytoremediation of Heavy-Metal-Contaminated Soils in Coal Mining Environments: A Systematic Review

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Abstract

Coal mining causes substantial soil, water, and environmental contamination. Phytoremediation represents a sustainable mitigation approach. This review systematically synthesizes peer-reviewed research on phytoremediation in coal-mining-impacted environments.

1. Introduction

Coal mining activities generate acid mine drainage and discharge heavy metal contaminants into surrounding terrestrial ecosystems. Environmental remediation is essential to restore soil quality and mitigate toxic metal bioaccumulation.

2. Research Gap

Existing chemical and physical soil remediation approaches have limitations related to high cost, severe site disturbance, and lack of long-term ecological sustainability. A systematic synthesis of phytoremediation performance in coal mine spoils is needed.

3. Review Purpose and Scope

To address these gaps, this review systematically synthesizes peer-reviewed literature published up to 2025 on phytoremediation strategies applied in coal mining environments.

4. Review Methodology

Literature was identified from major scientific databases including Web of Science, Scopus, and PubMed using search terms including coal mining, heavy metals, phytoremediation, and biochar. Studies published up to 2025 were evaluated.

5. Evidence Base and Techniques

Evidence was synthesized covering key mechanisms: Phytostabilization, Phytoextraction, Phytodegradation, Rhizofiltration, Microbe-assisted phytoremediation, and Biochar-assisted phytoremediation.

6. Key Findings and Synthesis

Heavy-metal-contaminated soils and acid mine drainage are identified as primary targets for phytoremediation. Effectiveness varies significantly with plant species, contaminant type, substrate conditions, and agronomic amendments.

7. Limitations

Existing evidence is constrained by short-term laboratory column experiments and a relative scarcity of long-term field-scale pilot validations in active mine sites.

8. Future Directions

Future research should prioritize long-term field validation, site-specific hyperaccumulator selection, standardized performance metrics, and safe biomass management.

References

Baker AJ (1981) Accumulators and excluders—strategies in the response of plants to heavy metals. J Plant Nutr 3:643-654.